

To further investigate the inter-relationship between PGAG and its parameter configuration, the patch size μ is set to 3, 5, and 7, while the adjacency ratio Ar varies over a range of values: 0.00005, 0.0001, 0.00025, 0.0005, 0.001, 0.0025, 0.005, 0.01, and 0.025. According to the recommendation in the manuscript, the step is set to $\Delta = (\mu - 1) / 2$. These settings constitute the parameter experiments, which are used to evaluate the fluctuations of κ on the Yellow River dataset. Here, we integrate PGAG and Otsu threshold for change detection to avoid the interference caused by the parameters of the classification method. The results are illustrated in Fig. 2, where each heatmap square denotes the value of κ and runtime under corresponding parameter settings. The data demonstrate that the κ values fluctuates minimally within each row, indicating that the proposed PGAG method is relatively insensitive to variations in the number of neighbor. Meanwhile, the κ in each row are characterized by an initial increase and subsequent decrease with the increment of parameter Ar . Therefore, it is suggested that an appropriate degree of connectivity between neighbors contributes to improved accuracy in change detection. From this experiments, it is evident that Ar values within the range of 0.001 to 0.01 are advantageous for the proposed PGAG method when applied to change detection tasks on the Yellow river dataset. Importantly, this range aligns well with the recommended parameter setting interval in our manuscript. In contrast, the κ values within each column vary considerably, indicating that the PGAG method is more sensitive to the size of patches. This is because, in the PGAG method, patches serve as the smallest units that constitute groups, affecting all aspects of the method, including group construction, PGAG construction and matching, as well as subsequent change measure and feedback. From the trend of time consumption, it can be observed that the runtime increases as the number of neighbors grows, but decreases as the step becomes larger. In terms of the numerical values of time consumption, the runtime is more significantly affected by the step than by the number of neighbors. This is because the step directly controls the number of vertices in the PGAG. For efficient execution of the algorithm, especially when dealing with large-scale SAR images, it is recommended to set a larger step. Based on the foregoing analysis, a patch size $\mu = 5$ ($\Delta = 2$) and an adjacency ratio $Ar = 0.01$ are commonly adopted as standard parameters in similar studies.

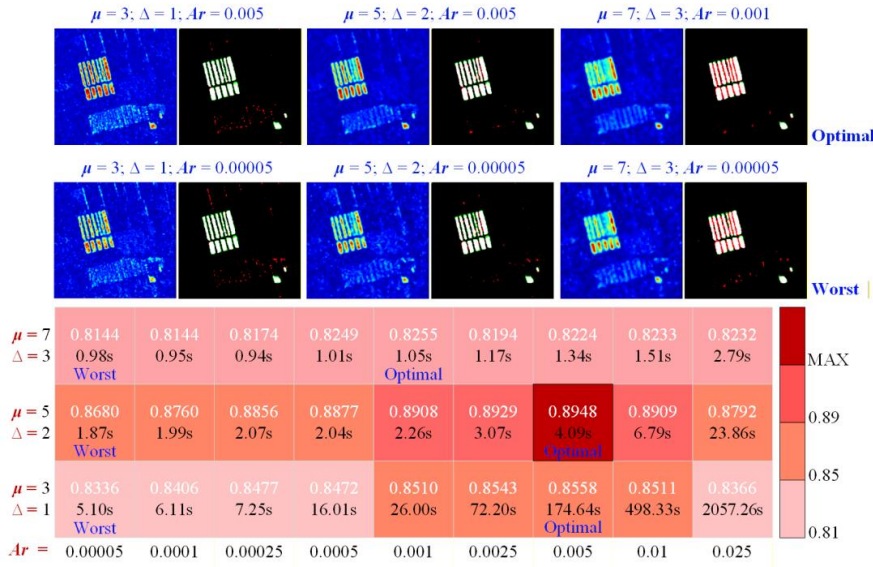


Fig 1. Parameters analysis for the proposed PGAG method on Yellow river dataset.